DIATraceExtractor V0.2: release report (2026-10-12)

What it is. `DIATraceExtractor` is an OpenMS TOPP tool (C++23, OpenMS develop) that extracts DIA fragment and precursor-isotope traces (XICs) from a predicted spectral library and writes them as Parquet/TSV. It calibrates retention time and mass against the run itself, needs no identifications, and does no scoring. This report gathers what has been measured about the quality of its traces and its speed against DIA-NN 1.7.12 and OpenSWATH, the design decisions those measurements led to, and what is still open. Details live in the dossier (`00-dossier.md`), the extraction description (`EXTRACTION.md`) and the backlog (`BACKLOG.md`).

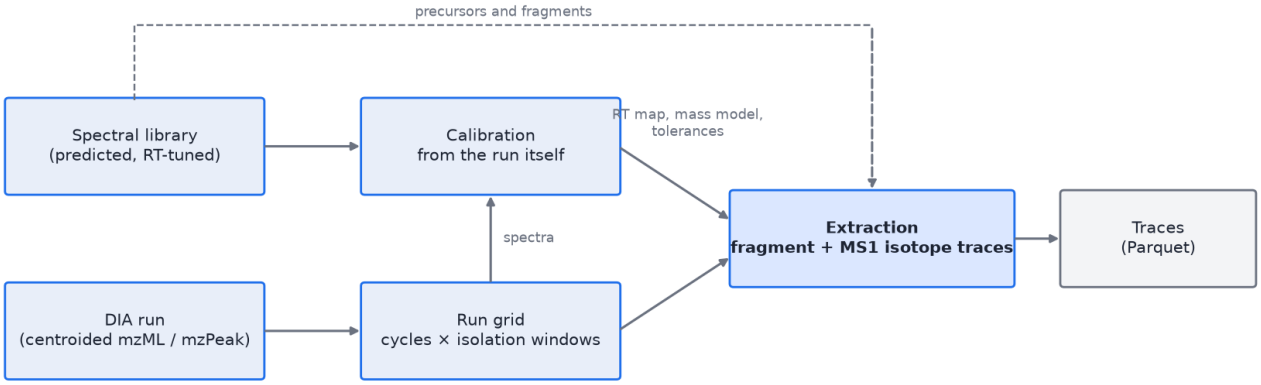
Verdict in one paragraph. At equal extraction rules the three tools produce the same traces; the differences between them are rule choices, and ours are the best-supported ones: a mass offset measured from the run (both instruments need it), a tolerance expressed in multiples of the measured error width, and a narrow, mass-corrected MS1 window that captures 98% of the signal at 1% interference where DIA-NN's export captures 92% at 5% and OpenSWATH 100% at 22%. The extraction runs 14–21× faster than DIA-NN and 40–100× faster than OpenSWATH on the same inputs, in 2.7 GiB, with byte-identical output at any thread count and from any of the three readers (fast mzML, OpenMS mzML, mzPeak). It runs end to end on a second instrument class (timsTOF diaPASEF) with the same agreement to DIA-NN as on the Orbitrap Astral. What remains open is one modelling question: the tolerance multiple is set per instrument class from the identification screen, not measured from the run, because the calibration anchors cannot see the interference that sets the optimum.

1. Data, libraries, references

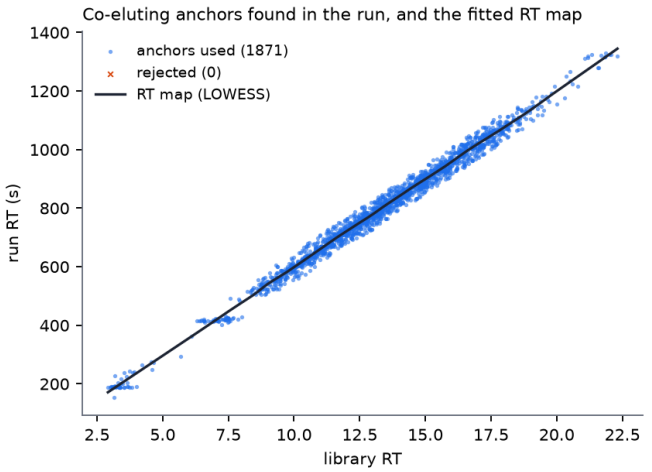
	Orbitrap Astral	**timsTOF SCP (diaPASEF)**
data set	ProteoBench DIA entrapment, 15 min, 50 ng, runs 01–03 (5.8 GB mzML each)	ProteoBench diaPASEF, conditions A and B, sample Alpha 01 (`.d`, converted to mzPeak and, by this tool, to mzML)
library	DIALibGen 0.11.1, human (3.5 M precursors) and entrapment (7.0 M); RT model tuned on DIA-NN IDs of run 01	DIALibGen, HYE mixed species, 4.15 M precursors (predicted RT, `-rt_mode auto`)
reference traces	DIA-NN 1.7.12 (last open-source version, `--xic` export patched), OpenSwathWorkflow (OpenMS develop)	DIA-NN 1.7.12 on our mzML export (1.7.12 reads no `.d`)
restriction for comparisons	DIA-NN's 1% FDR IDs (78–93 k precursors)	DIA-NN's IDs (54,557 A01; 55,665 B01)

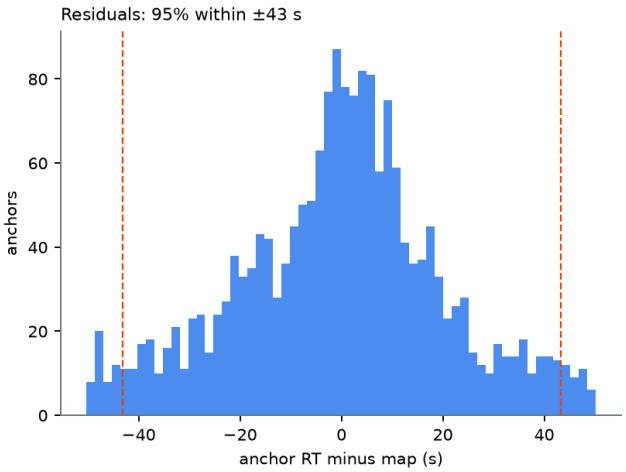
DIA-NN exports traces only while searching, so its times include the search; OpenSWATH's include its scoring.

2. How the extraction works

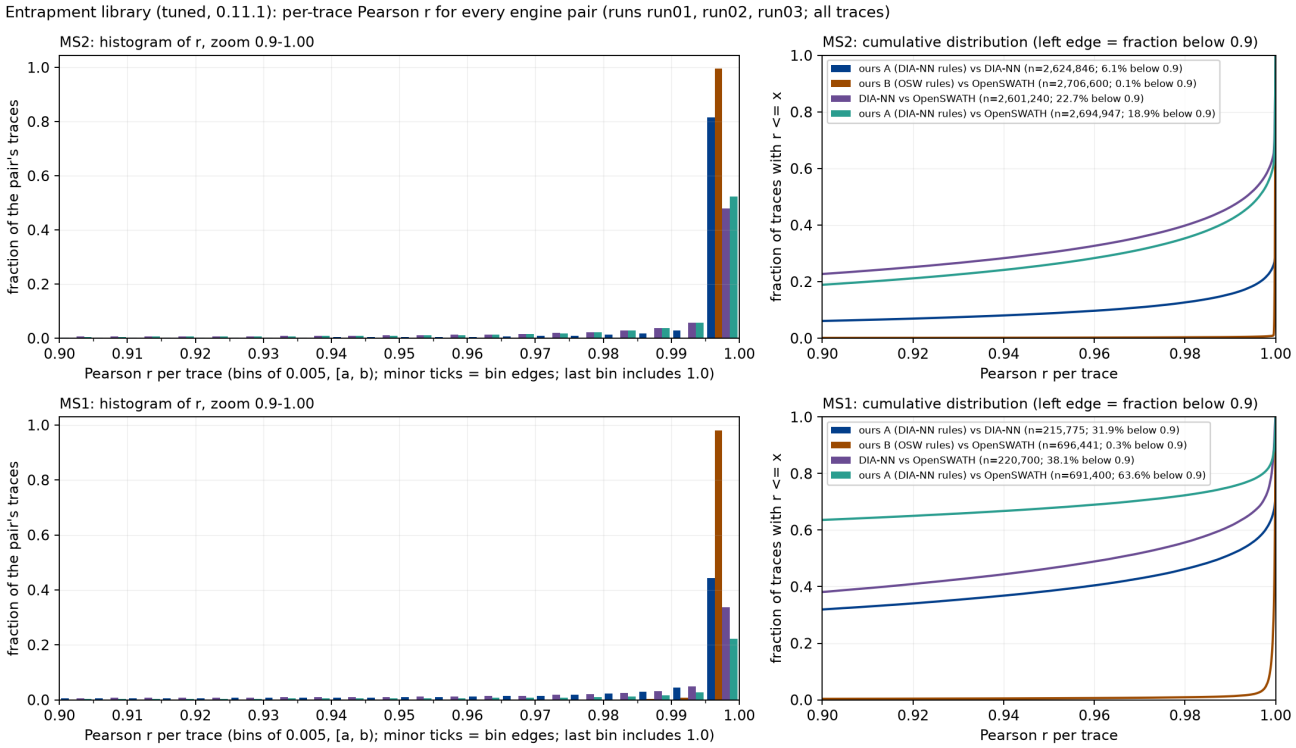


- **Run grid.** Every centroid of the run goes into one memory-mapped pool (m/z as uint32 at 1e-5 Th, intensity as float32: 8 bytes per peak), indexed by isolation window and DIA cycle, with MS1 on its own scan axis. Three readers feed it: a fast mzML parser, the OpenMS mzML reader, and the mzpeak-openms library for mzPeak files (diaPASEF frames are split per isolation window and collapsed across TIMS scans within 15 ppm).
- **Calibration** (`EXTRACTION.md` §3). A panel of 5% of the library (every k-th row, never the `-ids` subset) is searched for co-eluting fragment apexes: a strict first pass over the whole run yields anchors for a robust LOWESS RT map; a relaxed second pass within ±max(4σ, 30 s) of the map refines it. The apex centroids of the strict anchors give the mass model: a median offset, or a line offset + slope × m/z when the m/z dependence is material (new in V0.2), and the error width as 1.4826 × MAD of the residuals. Tolerances are multiples of that width.
- **Extraction.** Each precursor's fragments and MS1 isotopes are extracted within ±60 s of the mapped RT (or a measured window, `-rt_window 0`), by a merge join of the sorted target windows against each spectrum's centroids, taking the most intense centroid per window (DIA-NN's rule, `-aggregation max`) or their sum (OpenSWATH's, `sum`).
- **Scheduling.** One worker pool runs library reading, extraction, encoding and writing as task classes rebalanced by pipeline pressure, with in-order issue and an ordered commit, so the output does not depend on the thread count.





3. Are the traces the same? (Astral, shared tuned library, all DIA-NN IDs)



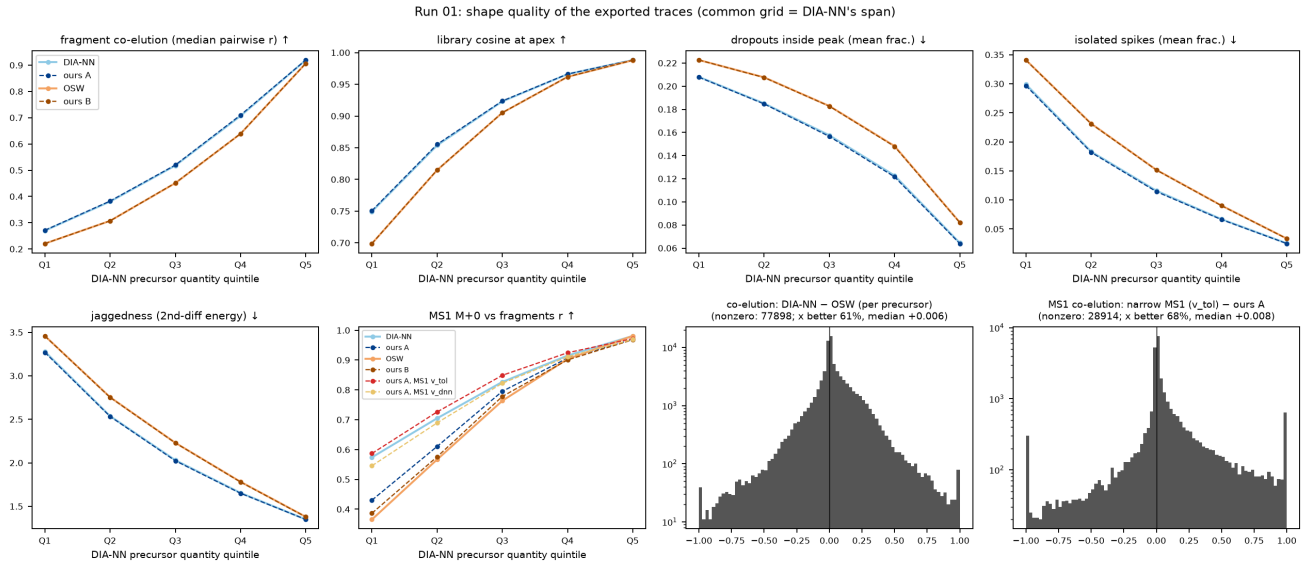
pair	MS2 median r	MS2 r ≥ 0.995	MS1 median r
ours B (OpenSWATH rules) vs OpenSWATH	0.9999	≈ 99%	0.9998
ours A (DIA-NN rules) vs DIA-NN	1.0000	81%	0.988
DIA-NN vs OpenSWATH	0.994	48%	0.964

Ours B reproduces OpenSWATH nearly exactly; ours A reproduces DIA-NN at MS2. The two reference engines disagree with each other more than either disagrees with its matching configuration of ours. The remaining MS2 tail against DIA-NN (3.5% of traces with r < 0.9 on run 01) is entirely its float32 quadratic window centre, shifted by about 0.5 ppm from ours: emulating that arithmetic reproduces every one of 35,000 DIA-NN traces in a sample (dossier §3.2). It is a DIA-NN rule choice, and its quadratic describes the true mass error worse than a constant does.

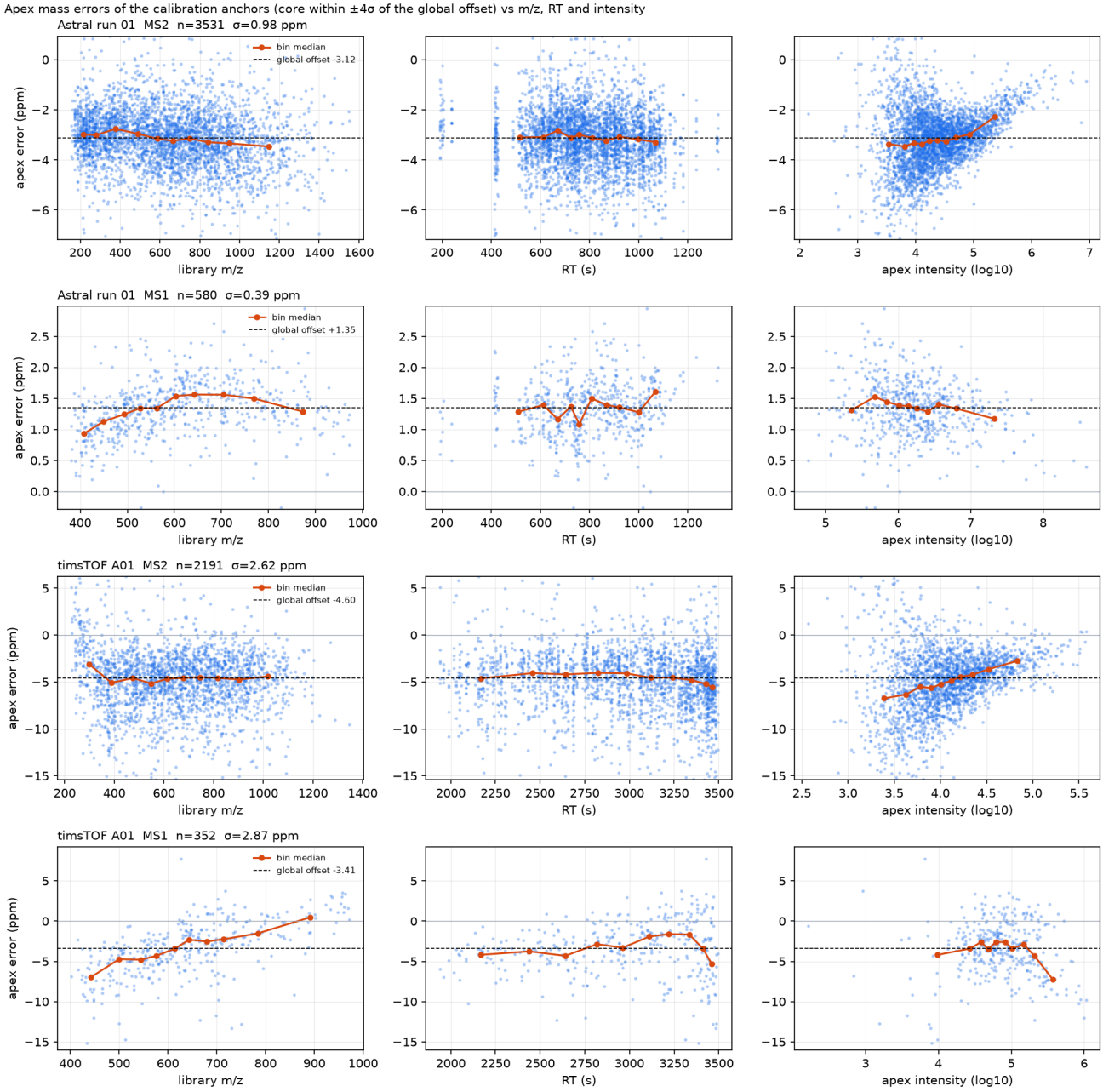
4. Which rules are right? (re-extraction from the raw centroids, 3,000 IDs)

	MS2	MS1
DIA-NN	99.5% capture, 2.9% interference	export at the **uncorrected** m/z: 92% capture, 5% interference
ours (V0.1 rule)	99.5% capture, 2.9% interference	narrow corrected window: **98% capture, 1.1% interference**
OpenSWATH	97.3% capture (no mass correction)	wide window: 100% capture, 22% interference

Shape metrics on the actual exports agree (run 01 / 02): fragment co-elution ours 0.541 / 0.560, DIA-NN 0.540 / 0.554, OpenSWATH 0.484 / 0.555; MS1-to-fragment co-elution ours 0.886 / 0.889, DIA-NN 0.880 / 0.895, OpenSWATH 0.826 / 0.802. OpenSWATH additionally places isotope k at m/z + k·1.00336 without dividing by the charge, so its M+1 and M+2 traces of charge ≥ 2 precursors are a different ion.


Run 01: shape quality of the exported traces (common grid = DIA-NN's span)
fragment co-elution (median pairwise r) ↑
library cosine at apex ↑
dropouts inside peak (mean frac.) ↓
isolated spikes (mean frac.) ↓
DIA-NN
ours A
OSW
ours B
DIA-NN precursor quantity quintile
jaggedness (2nd-diff energy) ↓
MS1 M+0 vs fragments r ↑
ours A, MS1 v_tol
ours A, MS1 v_dnn
co-elution: DIA-NN − OSW (per precursor)
(nonzero: 77898; x better 61%, median +0.006)
MS1 co-elution: narrow MS1 (v_tol) − ours A
(nonzero: 28914; x better 68%, median +0.008)

5. The structure of the mass errors, and the mass model



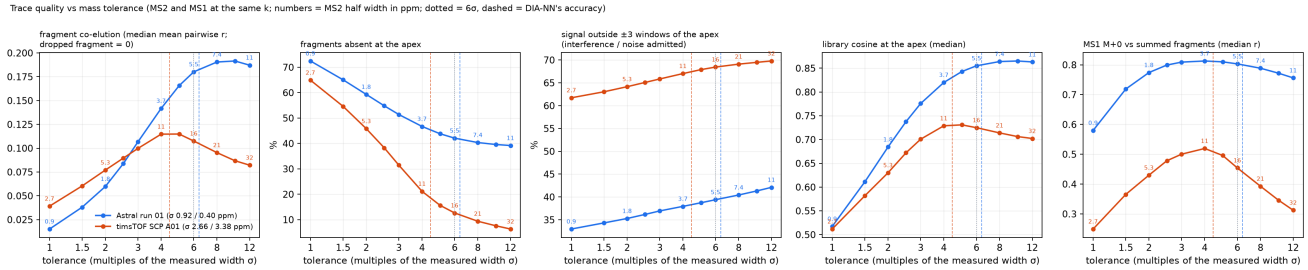
Apex errors of the calibration anchors against library m/z, RT and intensity (core within ±4σ of the offset):

	global offset	σ (MAD)	σ after an m/z + RT model	what the data show
Astral MS2	−3.12 ppm	0.98	0.94	centred after the offset; trends < 0.5 ppm
Astral MS1	+1.35 ppm	0.39	0.32	mild m/z curvature
timsTOF MS2	−4.60 ppm	2.62	2.55	small RT drift (−1.2 ppm over the run)
timsTOF MS1	−3.41 ppm	2.87	2.02	**linear in m/z: +15 ppm per 1000 Th**

- Both instruments have a systematic shift of 1–5 ppm; the offset is essential and the tool measures it.

- After the offset the residuals are centred (mean = median, no skew). An m/z-dependent term matters only for timsTOF MS1, where a line narrows σ by 29%. The mass model is therefore offset + slope × (m/z − reference), kept only when the shift over the central 90% of the m/z range exceeds half the width and three standard errors (`calibration:mass_model linear`, the default). On Astral it changes nothing measurable (agreement with DIA-NN: MS2 87.3%, MS1 72.3% vs 72.1% with the constant model on the shared run-01 library).
- The dominant remaining structure is **intensity**: weak centroids sit low, strong ones high (Astral MS2 −0.4 → +0.7 ppm, timsTOF MS2 −2.2 → +1.6 ppm from the weakest to the strongest decile). It cannot be corrected before extraction, and it is the reason the width measured on strong anchors understates what weak fragments need.

6. Trace quality as a function of the tolerance (both instruments)



The DIA-NN-identified precursors extracted at k × σ, k = 1 … 12, and scored on the traces (`scripts/tolerance_screen.py`):

	MS2 co-elution optimum	MS1 optimum	DIA-NN's own accuracy
Astral run 01	k = 8–10 (±7.4–9.2 ppm)	k = 4 (±1.6 ppm)	6.4σ (±5.9 ppm)
timsTOF A01	k = 4–5 (±10.6–13.3 ppm)	k = 4 (±13.5 ppm)	4.4σ (±11.8 ppm)

On Astral the curve saturates (heavy error tail, dropouts keep falling); on timsTOF it peaks and falls (Gaussian errors, interference costs beyond 5σ). DIA-NN's optimised accuracy sits at each instrument's optimum; MS1 peaks at 4σ on both.

Can the calibration find the multiple by itself? No. Every anchor-based criterion tried rises monotonically to 8σ on timsTOF too: the all-fragment co-elution curve of the anchors, the same on the weaker half of the anchors, on their weak fragments only, and on both; and the anchor yield at a 1% decoy-calibrated false rate, which the shifted-fragment decoys make meaningless (0–5 decoys per run). Anchors are present precursors whose fragments stay correlated as the window widens; the screen's optimum comes from interference in weak and absent precursors, a population the calibration does not have. **Decision:** the multiples are set per instrument class from the run's metadata (MS2 9σ Orbitrap/Astral, 4σ timsTOF and other TOF, 6σ with a warning when unknown; MS1 4σ), as explicit options (`-mz_sigma_ms2`, `-mz_sigma_ms1`; `-mz_tol*` for fixed ppm). A measured criterion needs a population containing absent peptides, i.e. peptide decoys from the library generator (backlog).

7. A second instrument: timsTOF SCP diaPASEF

Runs A01 and B01 go through the mzPeak reader (frames split per isolation window, TIMS scans collapsed within 15 ppm) and through DIA-NN 1.7.12 via our mzML export. Calibration finds 775 anchors (25 ng sample), MS2 width 2.6–2.7 ppm, MS1 2.2 ppm with the m/z model.

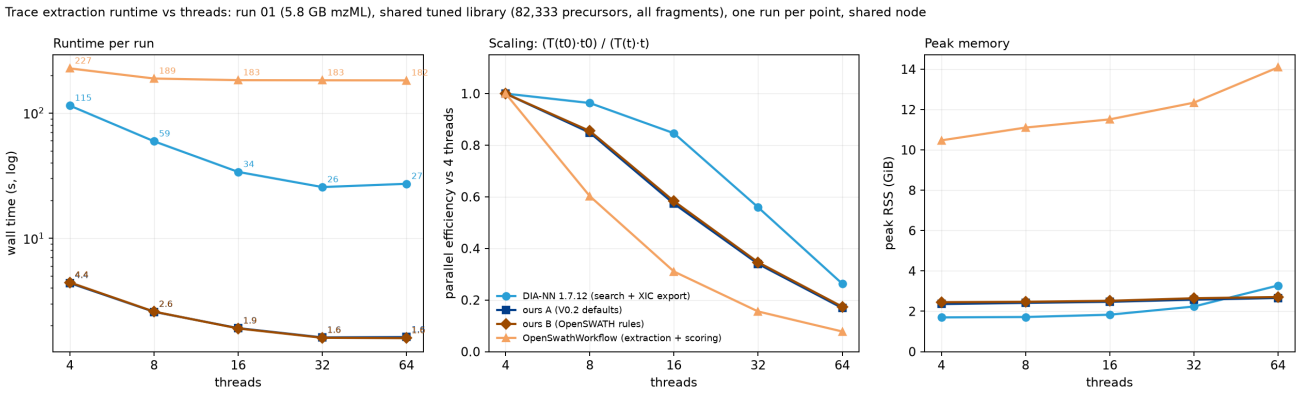
ours vs DIA-NN (its IDs)	MS2 median r	MS2 r > 0.9	MS1 r > 0.9
A01, defaults (4σ)	0.992	86.0%	63%
A01, at DIA-NN's ±11.8 ppm	0.9993	91.7%	60%
B01, defaults (4σ)	0.974	76.0%	67%
B01, at ±11.8 ppm (DIA-NN: ±12.6)	0.996	89.4%	66%

At equal tolerance the engines agree on diaPASEF as on Astral. **MS1 agreement of 60–67% looked like a defect and is not:** where both engines have signal the values are identical, and each has zeros where the other has signal, so the two windows sit at different centres. Placing our MS1 window at the uncorrected library m/z with DIA-NN's MS2 accuracy raises the agreement to 79% (widening to ±20 ppm drops it to 43%): DIA-NN's XIC export takes MS1 at the uncorrected m/z with its MS2 accuracy, the rule already found on Astral (§4), where the 1.3 ppm MS1 offset hides it; here the offset is 3.4 ppm plus the m/z slope. Our MS1 traces are the mass-corrected ones.

8. Performance

Machine: 2 × AMD EPYC 9654 (56 cores per socket online, SMT), shared node; inputs on local `/scratch`; one run per point unless stated; peak RSS from `/usr/bin/time`.

Runtime vs threads, run 01, shared tuned library (82,333 precursors, all fragments); ours at the V0.2 defaults (commit cb466f0), DIA-NN and OpenSWATH from the same series with unchanged binaries:



threads	ours A (defaults)	ours B (OpenSWATH rules)	DIA-NN 1.7.12	OpenSwathWorkflow
4	4.4 s / 2.4 GiB	4.4 s / 2.4 GiB	115 s / 1.7 GiB	227 s / 10.5 GiB
8	2.6 s	2.6 s	59 s	189 s
16	1.9 s	1.9 s	34 s	183 s
32	1.6 s	1.6 s	26 s	183 s
64	1.6 s / 2.7 GiB	1.6 s / 2.7 GiB	27 s / 3.3 GiB	182 s / 14.1 GiB

Ours is 21–26× faster than DIA-NN and 52–114× faster than OpenSWATH at every thread count (both references do more: DIA-NN searches, OpenSWATH scores). Ours flattens beyond 32 threads at this library size (1.6 s, of which ingest 0.3 s and calibration about 0.5 s); OpenSWATH stops scaling at 8 threads while its memory grows to 14 GiB. The node carried a load of 13–16 from other users during the series (`results/threads_cb466f0`).

Full libraries, run 01, 64 threads, top 6 fragments, V0.2 defaults:

library	total (2 runs)	calibration	extraction	peak RSS	traces
human, 3.49 M precursors	5.8–6.1 s	0.5 s	3.3–3.4 s	3.9 GiB	31.3 M
entrapment, 7.0 M precursors	9.3 s	1.0 s	6.1–6.2 s	4.8 GiB	63.0 M

Measured at a node load of about 21; the same binary measured 8.6–9.0 s on the entrapment library on a quieter node (dossier §4). Output is byte-identical for 1 and 64 threads and across the three readers.

diaPASEF (A01, 4.15 M-precursor library, 24 threads): 45 s wall, 13.8 GB peak RSS, 36.7 M traces; the mzPeak per-spectrum read and the mobility collapse take 20–26 s of it, the calibration 4 s, the extraction 12–13 s.

What the speed rests on (dossier §4, `06-parallel-efficiency.md`, `07-contention-diagnosis.md`): one mmap'd peak pool the parsers decode straight into; the library streamed in parts, never resident; one merge join per spectrum with a bucket index over m/z; a central scheduler with four task classes and an ordered commit; recycled trace and Arrow buffers with a per-thread cache; 32 Parquet parts written by their own tasks. Each step was accepted only with a byte-identical trace hash. The library-load transient (decoded row groups, parts and the joined copy coexisting) and the mzPeak reader's serialised decode (scales to 10–14 cores) are the known remaining costs.

9. Changes since V0.1

- Tolerances as multiples of the measured error widths, per instrument class (`-mz_sigma_ms2`, `-mz_sigma_ms1`); the instrument read from the run's metadata.
- Mass model with an m/z-dependent term when the data support it (`calibration:mass_model`).
- `-rt_window 0` = measured window; co-elution window from the measured peak width; calibration constants configurable; panel as a share of the library (`calibration:panel_fraction`, 5%).
- timsTOF diaPASEF: mzPeak frames split per window and collapsed across TIMS scans; `-out_mzml` exports the loaded run as centroided mzML; instrument model carried through the converter.
- mzPeak through the mzpeak-openms library (submodule, partitioned parallel read); Boost.JSON from the OpenMS vcpkg tree.
- Review leftovers closed: panel sampling for TSV/TraML/PQP, writer removes only its own shard names, atomic pool cache cap, identical centroid tie order across the three readers, overlapping windows counted.
- Libraries regenerated with DIALibGen's plain-string Parquet writer (timsTOF: 650 → 511 MB, panel load 3.6 → 1.3 s); DIALibGen adapted to current OpenMS develop (`OpenMS_CLI`).

10. Open issues (from `BACKLOG.md`)

- **Tolerance multiple as a measured criterion** needs peptide decoys from DIALibGen; the same decoys would replace the fixed anchor-acceptance thresholds (a decoy-calibrated cut exists, `calibration:anchor_fdr`, off by default because fragment-shift decoys are too easy).
- **RT window**: peak width and map uncertainty as a function of RT; iterative re-extraction with a narrowed window. RT model constants could be expressed in measured peak widths.
- **mzPeak reader** (other agent): decode scaling, the diaPASEF per-spectrum path, ions attached to the first precursor when `precursor_index` is NULL; **converter**: vendor vs ProteoWizard MS1 centroids as the reference.
- **Memory**: the library-load transient (~4.3 GiB on the 7 M library).
- **Untested**: overlapping or staggered isolation windows (counted and reported, nearest-window assignment or `-window_assignment all`); MS1-only records at the run's edges.
- Scheduler beyond 64 threads; compact-v2 library layout; the 50K shared library; results to the group.

11. Reproduction

- Build: `README.md` (OpenMS develop build with Arrow/Parquet from its vcpkg tree; `git submodule update --init`).
- Benchmarks: `scripts/benchmark_refined.sh` (three-way comparison on a shared recalibrated library), `scripts/bench_threads.sh` (thread series), `scripts/compare_traces.py` (pairwise metrics),

`scripts/tolerance_screen.py` and `plot_tolerance_screen.py` (§6), `scripts/plot_thread_scaling.py` (§8), `scripts/doc_figures.py` (§2 figures). Mass-error dumps: `ODIA_DUMP_MASS_ERRORS=file`.

- Inputs: ProteoBench DIA entrapment Astral module (https://proteobench.cubimed.rub.de/raws/DIA-astral-entrapment/) and diaPASEF module; libraries in `library/` (DIALibGen 0.11.1, `library/generate.sh`, `library/hye_timstof.log`).
- Tests: `ctest` in the build directory (7 tests: TraML extraction, OpenSWATH parity, byte-level Parquet compare).